

PRETORIA EENDRACHT, South Africa

WMO: 682620

Lat: 25.7426S

Lon: 28.1857E

Elev: 1308

StdP: 86.57

Time Zone: 2.00 (E02)

Period: 94-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.8	3.9	-10.7	1.8	15.2	-8.8	2.1	15.1	5.4	10.5	4.9	11.8	1.1	330	0.290

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.0	32.2	17.3	31.1	17.3	30.1	17.4	21.0	27.1	20.5	26.4	20.0	25.7	2.1	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	16.7	22.7	18.9	16.1	22.5	18.3	15.5	22.3	67.7	27.1	65.6	26.4	63.7	25.8	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.2	4.4	3.9	DB	0.4	35.2	1.3	1.4	-0.5	36.2	-1.3	37.0	-2.0	37.8	-3.0	38.9	(4)
			WB	-2.2	21.7	1.2	0.8	-3.0	22.3	-3.7	22.8	-4.4	23.3	-5.3	23.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.0	23.0	23.1	21.8	18.9	15.6	12.8	12.3	15.7	19.7	21.3	22.0	22.5
	DBStd	4.58	2.32	2.12	2.34	2.39	2.33	2.04	2.17	2.92	3.15	3.08	2.98	2.26
	HDD10.0	11	0	0	0	0	1	3	6	1	0	0	0	0
	HDD18.3	598	1	0	3	22	89	166	187	91	21	10	7	2
	CDD10.0	3307	403	368	365	267	173	88	77	178	291	349	361	388
	CDD18.3	854	146	135	110	38	4	0	0	10	62	101	117	132
	CDH23.3	5979	986	874	643	243	46	3	7	117	539	807	866	848
	CDH26.7	1691	298	259	164	42	3	0	1	13	148	263	276	224
Wind	WSAvg	1.5	1.6	1.4	1.4	1.2	1.1	1.2	1.3	1.6	1.8	1.9	1.9	1.7
Precipitation	PrecAvg	681	132	84	82	53	12	3	2	5	17	82	103	113
	PrecMax	948	503	213	193	147	62	23	25	26	75	227	217	202
	PrecMin	407	44	13	0	0	0	0	0	0	0	14	20	32
	PrecStd	129	83	47	51	45	17	7	5	6	17	53	55	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.4	33.1	32.2	30.0	27.2	23.6	24.1	28.3	32.1	33.3	33.3	32.2
		MCWB	18.9	18.5	17.8	16.2	14.1	11.9	10.6	13.0	14.7	15.6	16.8	18.3
	2%	DB	31.2	31.1	30.1	27.8	24.8	21.9	22.2	26.2	30.2	31.3	31.3	30.6
		MCWB	19.0	18.6	17.8	16.0	13.0	10.8	10.3	12.3	14.2	15.4	17.1	18.3
	5%	DB	29.8	29.7	28.5	26.1	23.2	20.3	20.8	24.5	28.3	29.7	29.8	29.1
		MCWB	18.9	18.7	17.6	15.7	12.7	10.3	10.0	11.9	13.5	15.5	17.1	18.5
	10%	DB	28.2	28.1	26.9	24.4	21.8	18.9	19.1	22.8	26.7	27.8	28.1	27.8
		MCWB	18.8	18.7	17.7	15.6	12.3	10.0	9.5	11.3	13.1	15.4	16.9	18.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.7	21.6	21.2	19.2	16.1	13.6	13.4	15.7	17.2	19.1	20.5	21.3
		MCDB	27.8	27.4	26.0	25.2	21.6	20.6	20.0	24.0	26.1	26.4	27.7	27.5
	2%	WB	21.0	20.8	20.3	18.0	15.2	12.6	12.1	14.4	16.1	18.1	19.6	20.5
		MCDB	27.6	26.5	25.3	23.1	20.7	19.1	18.5	22.5	25.4	25.8	26.3	26.6
	5%	WB	20.4	20.2	19.6	17.2	14.4	11.8	11.1	13.3	15.2	17.4	18.9	19.8
		MCDB	26.5	25.7	24.7	22.5	20.1	17.9	18.2	21.2	24.3	25.2	25.3	25.9
	10%	WB	19.8	19.7	18.9	16.5	13.6	11.0	10.3	12.3	14.4	16.8	18.3	19.3
		MCDB	25.5	25.0	24.1	22.0	19.3	16.7	17.2	20.1	23.2	24.3	24.7	25.2
Mean Daily Temperature Range	5% DB	MDBR	10.2	10.0	9.9	11.1	13.2	13.8	14.6	14.1	13.6	12.3	11.2	10.4
		MCDBR	12.7	12.4	12.6	13.6	15.1	15.9	16.8	16.2	15.7	14.9	13.7	12.4
		MCWBR	4.4	4.0	4.2	4.9	6.0	6.9	7.4	6.5	5.7	4.9	4.4	4.4
	5% WB	MCDBR	10.4	9.7	9.7	10.7	11.9	13.4	14.5	13.7	13.3	12.1	11.0	10.5
MCWBR		4.0	3.8	4.0	4.7	5.4	6.4	7.1	6.4	5.6	4.8	4.3	4.4	
Clear-Sky Solar Irradiance	taub	0.400	0.402	0.392	0.344	0.306	0.288	0.290	0.347	0.389	0.422	0.383	0.389	
	taud	2.314	2.306	2.313	2.414	2.489	2.516	2.498	2.332	2.233	2.179	2.332	2.335	
	Ebn at Noon	944	926	905	905	902	902	913	888	893	897	954	957	
	Edh at Noon	139	137	129	107	90	83	88	114	138	154	136	137	
All-Sky Solar Radiation	RadAvg	6.64	6.31	5.59	4.79	4.40	3.99	4.33	5.09	6.05	6.50	6.73	6.77	
	RadStd	0.68	0.56	0.52	0.43	0.22	0.20	0.11	0.23	0.28	0.43	0.62	0.40	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (12 neighbors)	+0.66	+0.75	N/A	N/A	N/A	N/A	N/A	-181	+234	+87

Nomenclature: See separate page